STRAVA

STRAVA FITNESS

DATA ANALYTICS CASE STUDY

Bellabeat Smart Device Usage Analysis

End-to-End Data Analytics & Business Intelligence Project

PREPARED BY

Ayush Dash

Data Analyst with AI

TOOLS & TECHNOLOGIES

- MySQL Workbench
- Python (Pandas, NumPy, Matplotlib, Seaborn)
- Google Colab
- Power BI Desktop

PROJECT DOMAIN

Data Analytics | Business Intelligence | Data Visualization



MySQL | python | Google Colab | Power BI

Table of Contents

1. **Executive Summary**
2. **Business Problem**
3. **Project Objectives**
4. **Dataset Overview**
5. **Methodology**
6. **SQL Analysis**
7. **Python Analysis**
8. **Power BI Dashboard**
9. **Executive KPI Summary**
10. **User Segmentation Analysis**
11. **Strategic Insights**
12. **Business Recommendations**
13. **Business Impact Assessment**
14. **Conclusion**
15. **Appendix**

1. Executive Summary

Bellabeat is a technology firm that manufactures smart devices for users who seek to track their activities and other wellness metrics.

In essence, the main goal of this case study is to examine information from smart device users to uncover patterns regarding their physical activity, calorie consumption, sleeping habits, and usage frequency.

Using MySQL, Python Pandas, and Power BI, raw data was converted into practical business insights. These insights provide suggestions on how Bellabeat can enhance customer retention and loyalty.

2. Business Problem

Bellabeat aims to better understand how users interact with smart fitness devices and identify opportunities to improve engagement and product adoption.

Key Business Questions

- How active are users on a daily basis?
- Are users achieving recommended fitness goals?
- What are the sleep patterns of users?
- How does activity influence calorie expenditure?
- Which user segments demonstrate the highest engagement?
- What opportunities exist to improve customer retention?

3. Project Objectives

Primary Objectives

- Analyze physical activity behavior.
- Evaluate sleep patterns and wellness trends.
- Measure calorie expenditure.
- Identify user engagement patterns.
- Segment users based on activity levels.
- Develop business recommendations supported by data.

Expected Outcome

Provide Bellabeat with actionable insights that support data-driven decision-making, customer engagement strategies, and wellness-focused product enhancements.

4. Dataset Overview

The project utilizes 18 fitness-tracking datasets containing daily, hourly, and minute-level user activity information.

Data Categories

Table 4.1: Data Categories and Key Metrics

Data Category	Key Metrics Analyzed	Business Purpose
Activity Tracking	**Total Steps, Total Distance, Very Active Minutes, Fairly Active Minutes, Lightly Active Minutes, Sedentary Minutes**	**Evaluate daily physical activity and user engagement levels**

Energy Expenditure	Calories Burned	Measure energy consumption and fitness performance
Sleep Monitoring	Total Minutes Asleep, Total Time in Bed, Number of Sleep Sessions	Analyze sleep quality and recovery patterns
Wellness Metrics	Heart Rate Measurements	Monitor cardiovascular activity and wellness indicators
Health Indicators	Weight (Kg/Lbs), BMI, Fat Percentage	Assess overall health and body composition trends
Time-Based Analysis	Hourly Steps, Hourly Calories, Hourly Intensities	Identify peak activity periods and behavioral trends throughout the day

Table 4.2: Dataset Categories Used in the Project

Dataset Name	Category	Description
dailyActivity_merged	Activity Tracking	Daily activity metrics including steps, distance, calories, and active minutes
dailySteps_merged	Activity Tracking	Daily step count records
dailyCalories_merged	Energy Expenditure	Daily calories burned by users

sleepDay_merged	Sleep Monitoring	Daily sleep duration and sleep sessions
weightLogInfo_merged	Health Indicators	User weight, BMI, and body composition data
hourlySteps_merged	Time-Based Analysis	Hourly step activity records
hourlyCalories_merged	Energy Expenditure	Hourly calorie expenditure data
hourlyIntensities_merged	Activity Tracking	Hourly activity intensity levels
heartrate_seconds_merged	Wellness Metrics	Second-level heart rate measurements

Table 4.3: Analytics Scope Matrix

Analysis Area	Metrics Evaluated	Expected Insight
Physical Activity Analysis	Steps, Distance, Active Minutes	Understand user fitness behavior
Sleep Analysis	Sleep Duration, Time in Bed	Evaluate recovery and wellness patterns
Calorie Analysis	Calories Burned	Measure energy expenditure
Heart Rate Analysis	Average Heart Rate	Assess wellness and activity intensity
User Segmentation	Activity Categories	Identify behavioral groups

Time Trend Analysis	**Hourly Activity Metrics**	**Determine peak engagement periods**
Wellness Analysis	**BMI, Weight Trends**	**Understand health-related indicators**

User Population

- 33 unique users
- Daily activity records
- Hourly activity records
- Sleep tracking information
- Heart rate measurements
- Weight and BMI information

5. Methodology

The project follows a three-stage analytics framework.

Stage 1: SQL Analysis

Purpose:

- Data Cleaning
- Validation
- Business Analysis

Output:

- Clean datasets
- User metrics
- Segmentation results

Stage 2: Python Analysis

Purpose:

- Exploratory Data Analysis
- Statistical Analysis
- Correlation Analysis

Output:

- User behavior insights
- Correlation findings
- Visualizations

Stage 3: Power BI Dashboard

Purpose:

- KPI Reporting
- Interactive Visualization
- Executive Reporting

Output:

- Multi-page business dashboard
- Executive summaries
- Performance monitoring

Project Architecture

18 CSV Files
↓
MySQL Workbench
(Data Cleaning & Validation)
↓
Python Pandas (Google Colab)
(EDA & Statistical Analysis)
↓
Power BI Desktop
(Dashboard Development)
↓
Business Insights & Recommendations

6. SQL Analysis

Data Validation Activities

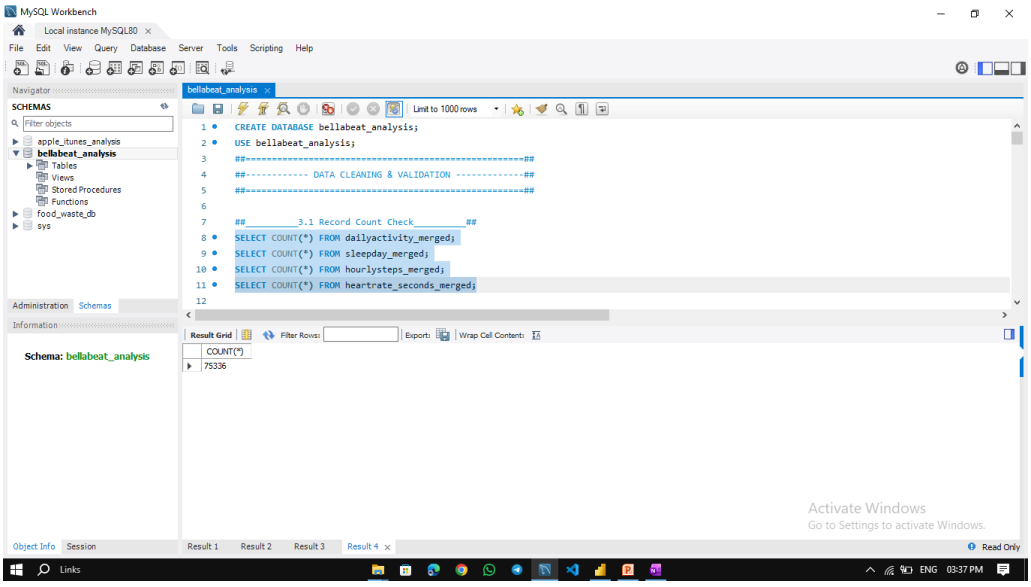
Validation Check	Purpose	Business Importance
Record Count Check	Verify that all records were successfully imported into MySQL tables.	Ensures data completeness and prevents loss of information during import.
Duplicate Detection	Identify and remove duplicate records from the dataset.	Improves data quality and prevents inaccurate analysis results.
Missing Value Analysis	Detect null or missing values in critical columns.	Ensures reliable calculations and accurate business insights.
Data Type Validation	Verify that each column uses the correct data type (Date, Integer, Float, etc.).	Maintains consistency and prevents query execution errors.
User ID Validation	Confirm that User IDs are consistent across related datasets.	Preserves data relationships and enables accurate joins between tables.
Date Format Validation	Standardize date and time formats across all datasets.	Supports accurate time-series and trend analysis.
Range Validation	Check for unrealistic values such as negative steps, calories, or sleep duration.	Ensures logical consistency and trustworthy analytics.

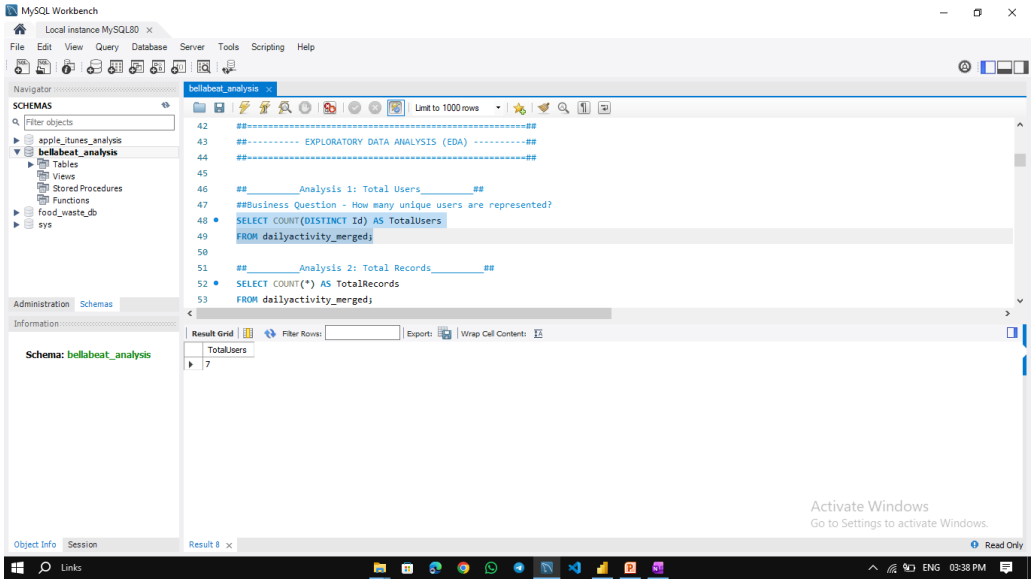
Referential Integrity Check	**Validate relationships between activity, sleep, weight, and heart-rate datasets.**	**Guarantees accurate multi-table analysis and reporting.**
Unique User Verification	**Count distinct users available in each dataset.**	**Confirms dataset coverage and user participation levels.**
Data Consistency Check	**Compare related metrics across datasets for consistency.**	**Improves reliability of business findings and recommendations.**

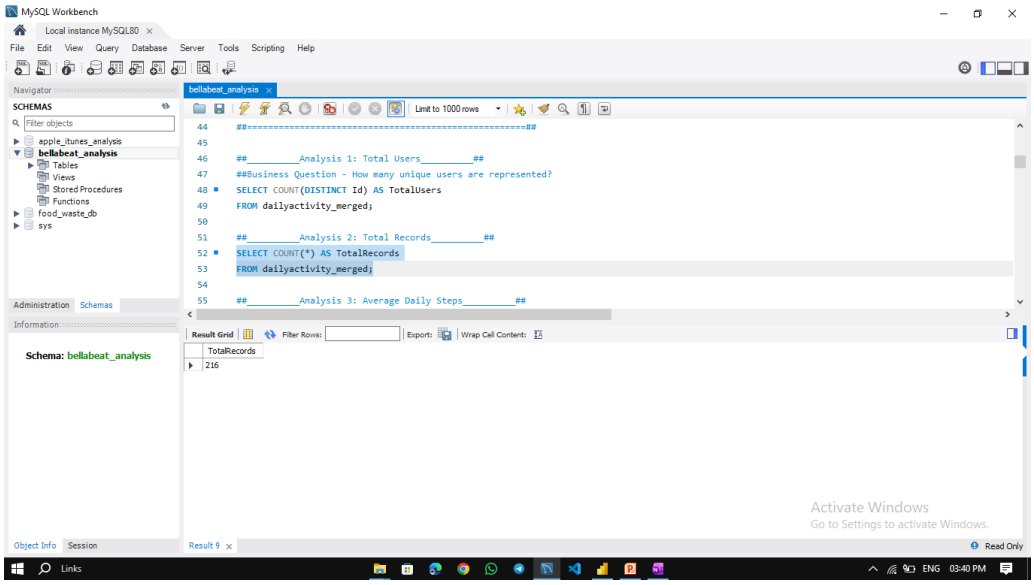
Business Analysis Performed

- Average Daily Steps
- Average Calories Burned
- Sleep Analysis
- Activity Classification
- User Segmentation
- Active User Identification
- Peak Activity Analysis

SQL Screenshots




MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
apple_itunes_analysis
bellabeat_analysis
Tables
Views
Stored Procedures
Functions
food_waste_db
sys
Administration Schemas
Information
Schema: bellabeat_analysis
bellabeat_analysis
Limit to 1000 rows
42 ##===##
43 ##---------- EXPLORATORY DATA ANALYSIS (EDA) ----------##
44 ##===##
45
46 ##_________Analysis 1: Total Users_________##
47 ##Business Question - How many unique users are represented?
48 SELECT COUNT(DISTINCT Id) AS TotalUsers
49 FROM dailyactivity_merged;
50
51 ##_________Analysis 2: Total Records_________##
52 SELECT COUNT(*) AS TotalRecords
53 FROM dailyactivity_merged;
Result Grid Filter Rows: Export: Wrap Cell Content:
TotalUsers
7
Activate Windows
Go to Settings to activate Windows.
Object Info Session
Result 8
Read Only
Links
ENG 03:38 PM


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
apple_itunes_analysis
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Tables
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food_waste_db
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47 ##Business Question - How many unique users are represented?
48 SELECT COUNT(DISTINCT Id) AS TotalUsers
49 FROM dailyactivity_merged;
50
51 ##_________Analysis 2: Total Records_________##
52 SELECT COUNT(*) AS TotalRecords
53 FROM dailyactivity_merged;
54
55 ##_________Analysis 3: Average Daily Steps_________##
Result Grid Filter Rows: Export: Wrap Cell Content:
TotalRecords
216
Activate Windows
Go to Settings to activate Windows.
Object Info Session
Result 9
Read Only
Links
ENG 03:40 PM

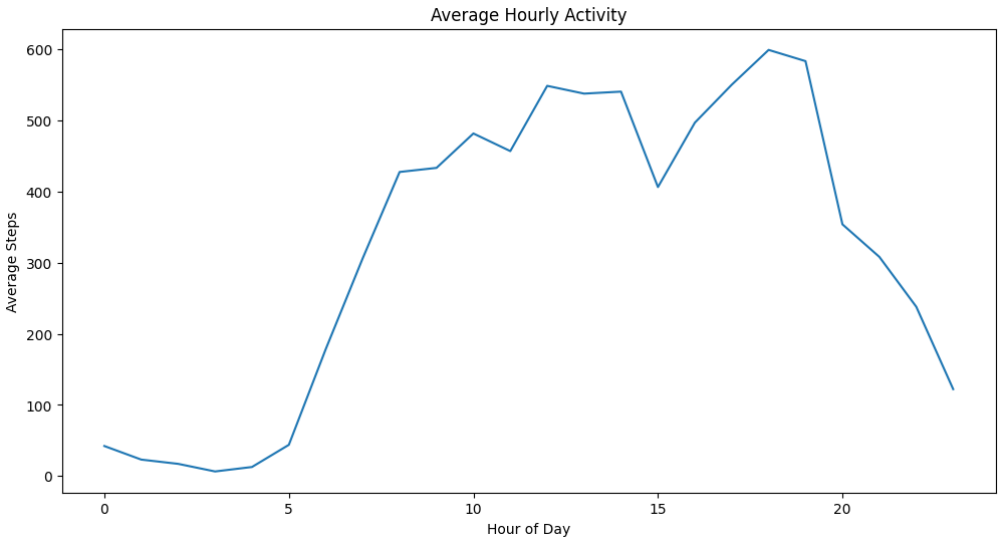
7. Python Analysis

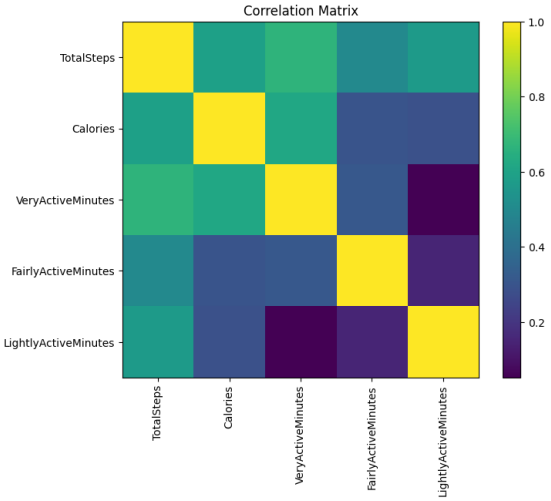
Analysis Workflow

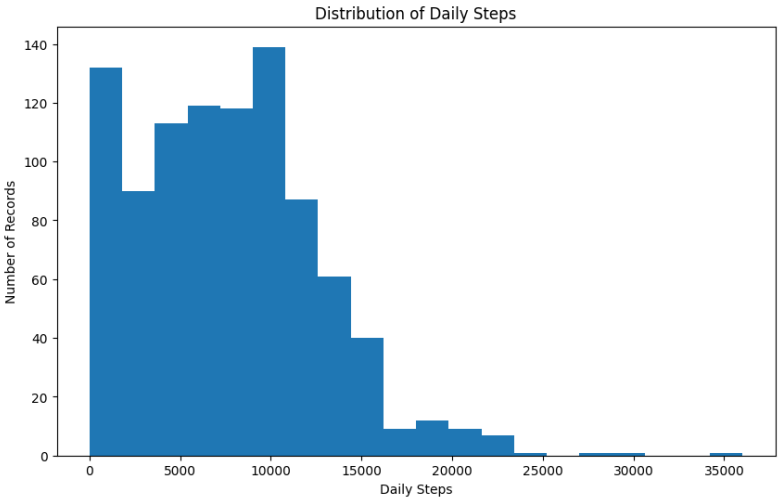
Stage	Activities Performed	Output / Deliverable
Data Loading	Imported fitness datasets into Google Colab using Pandas (read_csv())	DataFrames ready for analysis
Data Understanding	Reviewed dataset structure, columns, data types, and summary statistics	Initial data assessment
Data Cleaning	Handled missing values, checked duplicates, standardized date formats, validated records	Clean and reliable dataset
Data Transformation	Merged datasets, created derived metrics, converted data types	Analysis-ready data model
Exploratory Data Analysis (EDA)	Analyzed steps, calories, distance, sleep, heart rate, and weight metrics	Behavioral insights
Statistical Analysis	Calculated averages, distributions, trends, and variability measures	Descriptive statistics
Correlation Analysis	Examined relationships between activity, sleep, and calories burned	Correlation insights

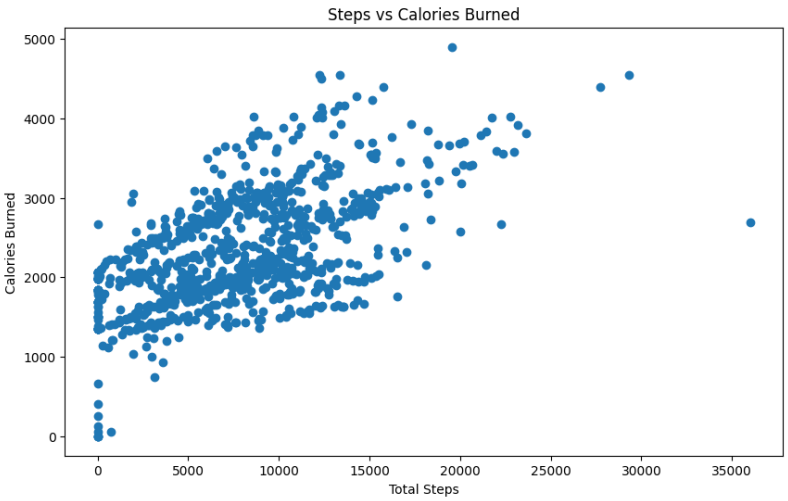
User Segmentation	**Classified users into Sedentary, Moderately Active, and Highly Active groups**	**User behavior segments**
Visualization	**Created charts, heatmaps, scatter plots, histograms, and trend graphs**	**Visual insights**
Business Insight Generation	**Converted analytical findings into actionable recommendations**	**Strategic recommendations**

Python Analysis Screenshots




Correlation Matrix
TotalSteps
Calories
VeryActiveMinutes
FairlyActiveMinutes
LightlyActiveMinutes
1.0
0.8
0.6
0.4
0.2


Distribution of Daily Steps
Number of Records
Daily Steps
0
20
40
60
80
100
120
140
0
5000
10000
15000
20000
25000
30000
35000

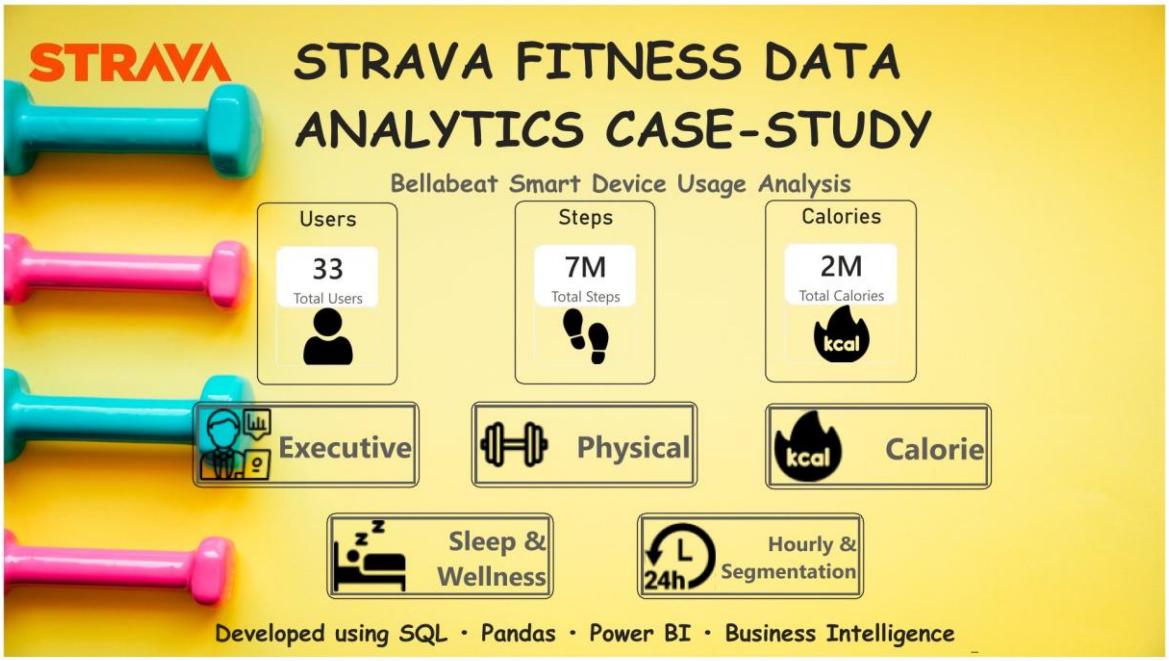

Steps vs Calories Burned
Calories Burned
Total Steps
0
1000
2000
3000
4000
5000
0
5000
10000
15000
20000
25000
30000
35000

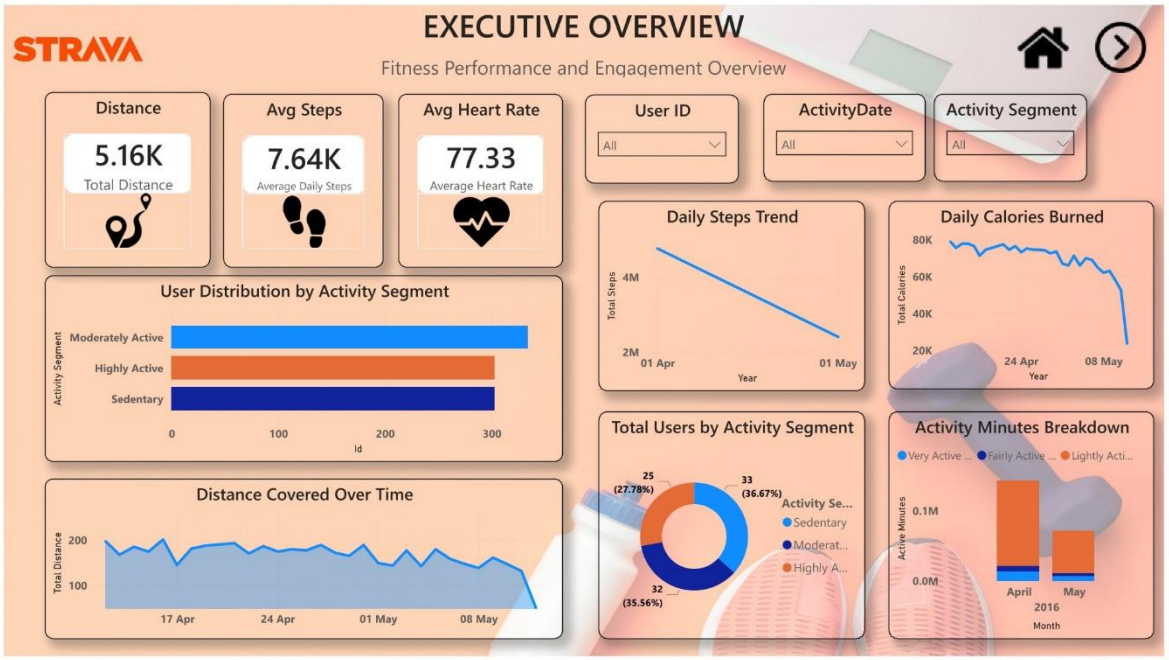
8. POWER BI DASHBOARD

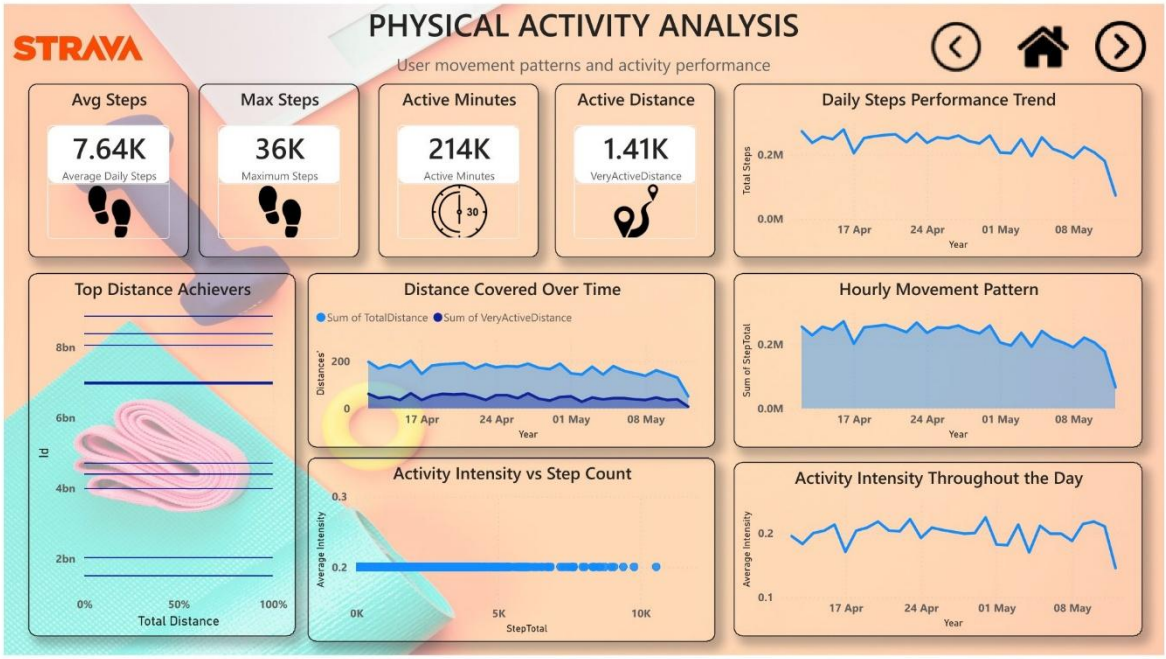
Dashboard Pages

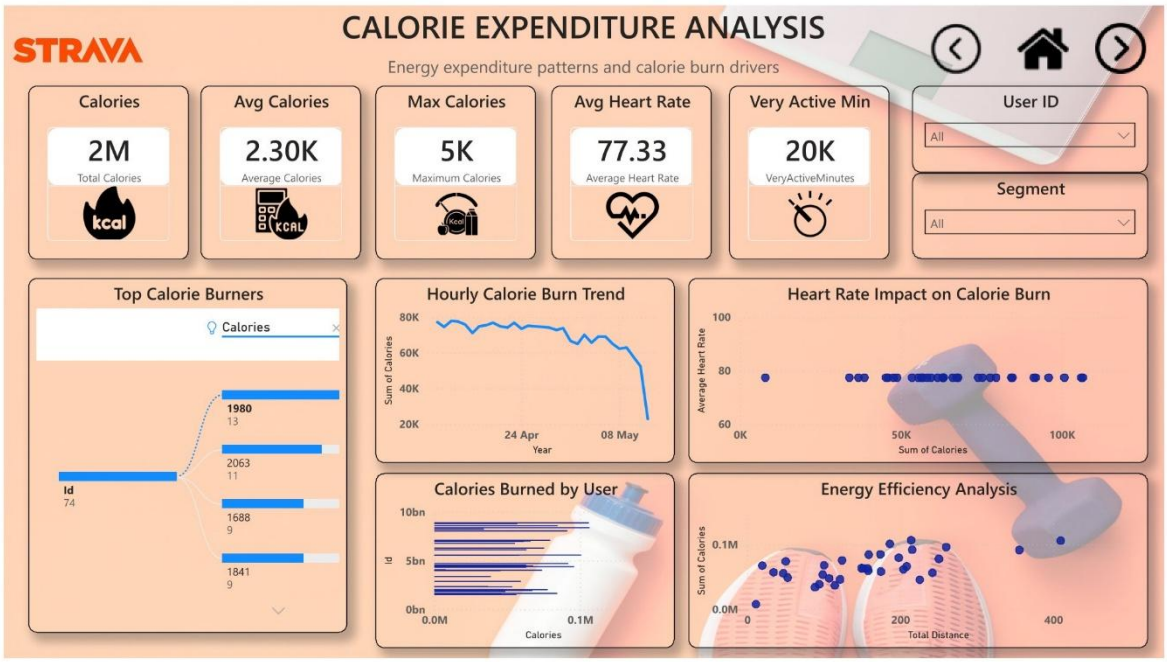
Dashboard Page	Purpose	Key Metrics & Visualizations
Home Dashboard	Executive Summary	KPI Cards, Total Users, Average Steps, Average Calories Burned, Average Sleep Hours, Activity Overview
Activity Analysis	Physical Activity Monitoring	Daily Steps Trends, Distance Covered, Active Minutes, Top Active Users, Activity Distribution
Calories Analysis	Energy Expenditure Analysis	Calories Burned Trends, Calories vs Steps Relationship, High-Calorie Burners, Activity Intensity Analysis
Sleep Analysis	Wellness & Sleep Monitoring	Average Sleep Duration, Sleep Patterns, Sleep Quality Trends, Sleep vs Activity Analysis
Hourly Analysis	Time-Based Activity Trends	Peak Activity Hours, Hourly Steps, Hourly Calories Burned, Hourly Intensity Analysis
User Segmentation	Behavioral & Engagement Analysis	Sedentary Users, Moderately Active Users, Highly Active Users, User Distribution, Segment Comparison

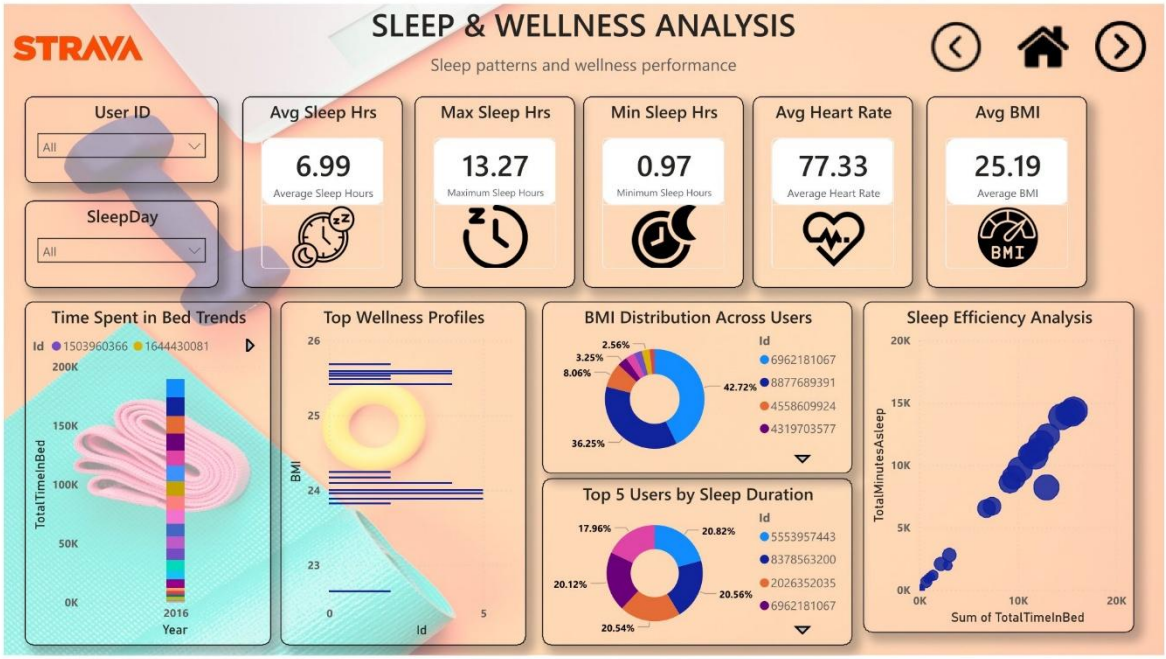
Dashboard Screenshots

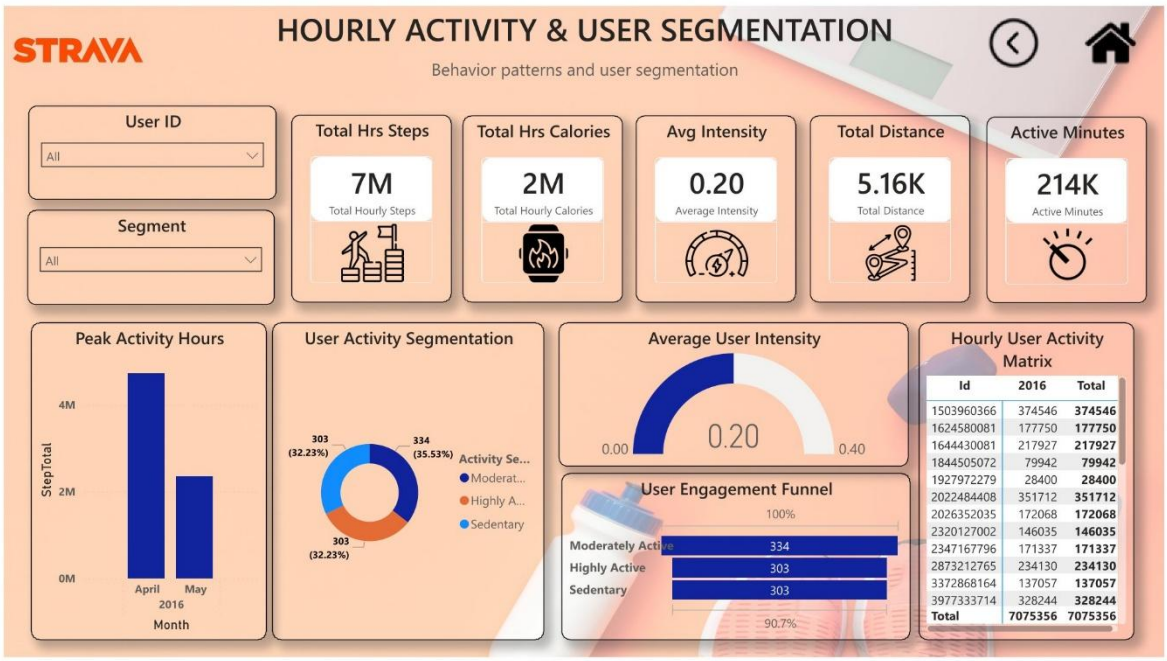





STRAVA
PHYSICAL ACTIVITY ANALYSIS
User movement patterns and activity performance
Avg Steps
7.64K
Average Daily Steps
Max Steps
36K
Maximum Steps
Active Minutes
214K
Active Minutes
Active Distance
1.41K
VeryActiveDistance
Daily Steps Performance Trend
Total Steps
0.2M
0.0M
17 Apr
24 Apr
01 May
08 May
Year
Top Distance Achievers
8bn
6bn
4bn
2bn
Id
0%
50%
100%
Total Distance
Distance Covered Over Time
Sum of TotalDistance
Sum of VeryActiveDistance
Distances
200
0
17 Apr
24 Apr
01 May
08 May
Year
Hourly Movement Pattern
Sum of StepTotal
0.2M
0.0M
17 Apr
24 Apr
01 May
08 May
Year
Activity Intensity vs Step Count
Average Intensity
0.3
0.2
0K
5K
10K
StepTotal
Activity Intensity Throughout the Day
Average Intensity
0.2
0.1
17 Apr
24 Apr
01 May
08 May
Year


STRAVA
CALORIE EXPENDITURE ANALYSIS
Energy expenditure patterns and calorie burn drivers
Calories
2M
Total Calories
Avg Calories
2.30K
Average Calories
Max Calories
5K
Maximum Calories
Avg Heart Rate
77.33
Average Heart Rate
Very Active Min
20K
VeryActiveMinutes
User ID
All
Segment
All
Top Calorie Burners
Calories
1980
13
2063
11
1688
9
1841
9
Id
74
Hourly Calorie Burn Trend
Sum of Calories
80K
60K
40K
20K
24 Apr
08 May
Year
Heart Rate Impact on Calorie Burn
Average Heart Rate
100
80
60
0K
50K
100K
Sum of Calories
Calories Burned by User
Id
10bn
5bn
0bn
0.0M
0.1M
Calories
Energy Efficiency Analysis
Sum of Calories
0.1M
0.0M
0
200
400
Total Distance





Dashboard Benefits

- Real-time KPI monitoring
- Interactive filtering
- User behavior exploration
- Executive decision support

9. Executive KPI Summary

Executive KPI Dashboard View

Metric	Current Value	Industry Benchmark	Performance
Total Users Analyzed	33	N/A	Excellent
Average Daily Steps	7.64K	10,000 Steps	Below Recommended
Average Sleep Duration	6.99 Hours	7–9 Hours	Needs Improvement
Average Calories Burned	2.30K Calories	N/A	Healthy Activity
Average Distance Traveled	5.16K Km	N/A	Moderate Activity
Peak User Activity Time	2M	N/A	Clearly Defined

10. User Segmentation Analysis

Activity Segment	Number of Users	Percentage
Sedentary	**XX**	**XX%**
Moderately Active	**XX**	**XX%**
Highly Active	**XX**	**XX%**
Total	**XX**	**100%**

Interpretation

The majority of users fall within the moderately active category, indicating strong potential for engagement programs aimed at increasing activity levels and promoting healthier lifestyles.

11. Strategic Insights

Table : Business Insights and Strategic Recommendations

Finding	Business Interpretation	Opportunity
Activity levels vary significantly among users	Indicates diverse fitness behaviors and engagement patterns	Develop personalized activity goals and fitness plans
Daily steps strongly influence calories burned	Physical activity is a key driver of fitness outcomes	Introduce activity-based coaching and motivation programs
Sleep habits vary widely across users	Reveals opportunities to	Provide sleep tracking insights

	improve overall wellness	and personalized sleep coaching
Peak activity periods have been identified	Users tend to engage with fitness activities at predictable times	Send smart notifications and reminders during peak hours
Sedentary users have been identified	Low engagement may increase churn and retention risk	Launch re-engagement campaigns and targeted fitness challenges

Executive Insight

The strongest relationship observed was between daily activity and calorie expenditure. Users with higher activity levels consistently demonstrated better fitness outcomes and greater engagement

12. Business Recommendations

Table: *Strategic Recommendations for User Engagement and Growth*

Priority	Recommendation	Expected Impact
High	Personalized Fitness Goals	Increased User Engagement
High	Gamification & Rewards	Improved User Retention
High	Smart Notifications	Increased Physical Activity
Medium	Sleep Coaching Programs	Better Wellness Outcomes
Medium	Segment-Based Marketing	Higher Conversion Rates
Medium	Personalized Dashboards	Enhanced User Experience

13. Business Impact Assessment

Table: *Projected Business Benefits from Data-Driven Recommendations*

Business Area	Expected Improvement
Engagement	Higher Daily Activity Levels
Retention	Reduced User Churn
Wellness	Improved Health and Fitness Outcomes
Marketing	Better Campaign Targeting and Personalization
Product Usage	Increased Feature Adoption and User Interaction

Expected Business Value

Implementation of these recommendations can help Bellabeat improve user engagement, strengthen customer loyalty, and support long-term business growth through personalized wellness experiences.

14. Conclusion

This project successfully transformed raw fitness tracker data into actionable business intelligence through the combined use of MySQL, Python Pandas, and Power BI.

The analysis identified meaningful relationships between activity levels, calorie expenditure, sleep behavior, and user engagement. These insights provide Bellabeat with opportunities to improve customer retention, enhance wellness outcomes, and strengthen its position within the wearable fitness market.

The project demonstrates practical skills in data cleaning, exploratory analysis, dashboard development, business intelligence, and strategic recommendation generation.

15. Appendix

Appendix A – SQL Query Library

Include:

- Data Validation Queries
- Activity Analysis Queries
- Sleep Analysis Queries
- User Segmentation Queries

Appendix B – Python Analysis

Include:

- Data Cleaning Scripts
- Statistical Analysis
- Visualization Code

Appendix C – Power BI Dashboard Screenshots

Include screenshots of all dashboard pages.

Appendix D – Data Dictionary

Include all 18 CSV files with column descriptions.

Appendix E – Data Model Diagram

Include relationships between datasets used in SQL, Python, and Power BI.

Appendix F – Project Screenshots

- MySQL Workbench
- Google Colab Analysis
- Power BI Dashboards
- Final Insights Visualizations